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Robot

Abstract

A robot includes a base; a first arm rotatably connected to the base around a first rotation axis; and a second arm rotatably connected to the first arm around a second rotation axis parallel to the first rotation axis, wherein the second arm includes an arm base and a cover configured to attach to and detach from the arm base and the cover includes a cover main body that covers an upper portion of the arm base and a covering section that protrudes downward from the cover main body and that covers a portion of a side surface of the arm base.

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Background/Summary

[0001] The present application is a continuation of U.S. patent application Ser. No. 18/517,491 filed Nov. 22, 2023, which is based on, and claims priority from JP Application Serial Number 2022-187720, filed Nov. 24, 2022, the disclosures of which are hereby incorporated by reference herein in their entireties.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a robot.

2. Related Art

[0003] A robot described in JP-A-2011-093066 is provided with a base, a first arm rotatably connected to the base around a first rotation axis, a second arm rotatably connected to the first arm around a second rotation axis parallel to the first rotation axis, and a working head supported by the second arm. In addition, the second arm includes an arm base, a joint actuator which is installed on the arm base and is configured by a motor, a decelerator, and the like, and a cover which is detachably and attachably mounted on the arm base. The cover is mounted on the arm base so as to cover the joint actuator and other drive systems installed on the arm base. When the joint actuator and other drive systems installed on the arm base of the second arm are inspected or repaired, an operation of detaching the cover from the arm base is performed.

[0004] However, in the robot described in JP-A-2011-093066, it is difficult to recognize the boundary between the arm base and the cover. Therefore, there is a problem in that it is difficult for an operator to recognize which portion should be held when the operator performs the operation of detaching the cover from the arm base, and a detachment operation of the cover cannot be easily performed.

SUMMARY

[0005] A robot according to the present disclosure includes a base; a first arm rotatably connected to the base around a first rotation axis; and a second arm rotatably connected to the first arm around a second rotation axis parallel to the first rotation axis, wherein the second arm includes an arm base and a cover configured to attach to and detach from the arm base and the cover includes a cover main body that covers an upper portion of the arm base and a covering section that protrudes downward from the cover main body and that covers a portion of a side surface of the arm base.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is an overall view of a robot system according to a first embodiment.

[0007] FIG. 2 is a side view of a second arm of a robot shown in FIG. 1.

[0008] FIG. 3 is a view as viewed from the direction of the arrow A in FIG. 2.

[0009] FIG. 4 is a view as viewed from the direction of the arrow C in FIG. 3.

[0010] FIG. 5 is a cross-sectional view taken along the line B-B in FIG. 3.

[0011] FIG. 6 is a view for explaining a procedure for detaching the cover from the arm base, and is a view of the second arm as viewed from above.

[0012] FIG. 7 is a view for explaining the procedure for detaching the cover from the arm base, and is a view of the second arm as viewed from above.

[0013] FIG. 8 is a view for explaining the procedure for detaching the cover from the arm base, and

is a view of the second arm as viewed from above.

DESCRIPTION OF EMBODIMENTS

[0014] Hereinafter, a robot according to the present disclosure will be described in detail based on embodiments shown in the accompanying drawings.

First Embodiment

[0015] FIG. 1 is an overall view of a robot system according to a first embodiment. FIG. 2 is a side view of a second arm of a robot shown in FIG. 1. FIG. 3 is a view as viewed from the direction of the arrow A in FIG. 2. FIG. 4 is a view as viewed from the direction of the arrow C in FIG. 3. FIG. 5 is a cross-sectional view taken along the line B-B in FIG. 3. FIGS. 6 to 8 are views for explaining a procedure for detaching the cover from the arm base, and are views of the second arm as viewed from above.

[0016] In FIG. 1, an X-axis, a Y-axis, and a Z-axis which are three axes orthogonal to each other are set, and the same three axes are shown in FIGS. 2, 3, 4, 5, 6, 7, and 8. Among the three axes, a Z-axis direction indicates a vertical direction, and an X-Y plane indicates a horizontal plane.

[0017] The up-down direction in FIGS. 1, 2, 3, and 5 coincides with the vertical direction. An upper side and a lower side in FIGS. 1, 2, 3, and 5 are also referred to as “upper” and “lower”, respectively, and for a robot arm 22, a first arm 23, a second arm 24, and the like, the right side in FIGS. 1 and 2, that is, an end portion close to a base 21 is referred to as a base end portion, and the left side, that is, an end portion close to a working head 25 is referred to as a tip end portion. In addition, as shown in FIG. 1, a posture in which the base 21, the first arm 23, and the second arm 24 are positioned on a straight line, that is, positioned along the X-axis is referred to as an initial posture.

[0018] In the present specification, the term “vertical” means not only a case where coincident with the vertical, but also a case where slightly inclined with respect to the vertical, for example, within $+10^\circ$. In the present specification, “parallel” means not only a case where two objects are parallel to each other, but also a case where two objects are slightly inclined, for example, within $+10^\circ$ from parallel to each other.

[0019] A robot system 1 shown in FIG. 1 includes a robot 2 and a robot control device 9 for controlling the driving of the robot 2.

[0020] The robot 2 in the present embodiment is a SCARA robot, and is used, for example, operations such as holding, transporting, assembling, and inspecting a workpiece such as an electronic component. However, the use of the robot 2 is not particularly limited. The robot according to the disclosure may be, for example, a six axes articulated robot, a dual-arm robot, or the like other than the SCARA robot.

[0021] As shown in FIG. 1, the robot 2 includes the base 21 and the robot arm 22 rotatably connected to the base 21. The base 21 is fixed to a floor surface 100 parallel to the horizontal plane. The robot control device 9 is installed inside the base 21. Unlike the shown configuration, the robot control device 9 may be installed outside the base 21.

[0022] The robot arm 22 includes the first arm 23 whose base end portion is connected to the base 21 and which rotates around a first rotation axis J1 along the vertical direction with respect to the base 21, and the second arm 24 whose base end portion is connected to a tip end portion of the first arm 23 and which rotates around a second rotation axis J2 along the vertical direction with respect to the first arm 23.

[0023] The working head 25 is provided at a tip end portion of the second arm 24. The second arm 24 has a shape extending linearly, that is, in the X-axis direction from the second rotation axis J2 to the working head 25. The working head 25 includes a spline nut 251 and a ball screw nut 252 disposed coaxial with each other at the tip end section of the second arm 24, and a spline shaft 253 that is inserted through the spline nut 251 and the ball screw nut 252. The spline shaft 253 is rotatable with respect to the second arm 24 around a third rotation axis J3, which is a central axis of the second arm 24 and extends along the vertical direction, and is movable in the up-down

direction along the third rotation axis J3.

[0024] The base **21** and the second arm **24** are connected to each other by a flexible pipe **10** such as a cable. A base end portion of the pipe **10** penetrates an upper surface **210** of the base **21** and is positioned in the base **21**. A tip end portion of the pipe **10** is positioned inside a cover **4**, and penetrates a wide opening section **412** (to be described later) formed in a top surface **41A**, which is an upper surface of the second arm **24**.

[0025] A power supply line, a signal line, and the like extending from the inside of the base **21** to the inside of the second arm **24** are accommodated inside the pipe **10**. That is, the power supply line, the signal line, and the like are configured as an insulation-coated cord or a cable obtained by bundling cords. The robot control device **9** installed inside the base **21** and a second joint actuator **28**, a first drive mechanism **291**, and a second drive mechanism **292** positioned inside the cover **4** are electrically connected to each other by such a cord or cable.

[0026] Since the pipe **10** has flexibility, the pipe **10** can sufficiently follow changes in the posture of the robot arm **22** by rotation of the first arm **23** and the rotation of the second arm **24**.

[0027] An end effector (not shown) is attached to a lower end portion **26** of the spline shaft **253**. The end effector is detachable and attachable, and one suitable for the target operation is appropriately selected. Examples of the end effector include a hand or a chuck that grips a workpiece which is an object to be assembled or processed, a tool that performs drilling, grinding, polishing, or the like, and a coating tool such as a spray gun.

[0028] The robot **2** includes a first joint actuator **27** that couples the base **21** and the first arm **23** and rotates the first arm **23** around the first rotation axis J1 with respect to the base **21** and the second joint actuator **28** as a joint section that couples the first arm **23** and the second arm **24** and rotates the second arm **24** around the second rotation axis J2 with respect to the first arm **23**.

[0029] The first arm **23** is connected to the base **21** in a state of being separated from the upper surface **210** of the base **21** by the first joint actuator **27**. The second arm **24** is connected to the first arm **23** in a state of being separated from an upper surface **230** of the first arm **23** by a predetermined distance by the second joint actuator **28**. The separation distance between a lower end of the second arm **24** and the upper surface **230** is not particularly limited, but is desirably set to such an extent that a finger F of a right hand HR or a left hand HL can enter between them during a detachment operation of the cover **4** (to be described later).

[0030] The robot **2** includes the first drive mechanism **291**, which rotates the spline nut **251** to rotate the spline shaft **253** around the third rotation axis J3, and the second drive mechanism **292**, which rotates the ball screw nut **252** to move up and down the spline shaft **253** in a direction along the third rotation axis J3. The spline shaft **253** is rotated in a predetermined direction around the third rotation axis J3 by the operation of the first drive mechanism **291**, and the lower end portion **26** is rotated in the same direction in accordance with the rotation. The spline shaft **253** is moved in the axial direction of the third rotation axis J3 by the operation of the second drive mechanism **292**, and the lower end portion **26** is accordingly moved in the same direction, that is, raised or lowered.

[0031] Each of the first joint actuator **27**, the second joint actuator **28**, the first drive mechanism **291**, and the second drive mechanism **292** includes a motor, a decelerator, an encoder, and the like (not shown). Each motor and each encoder are electrically connected to the robot control device **9**. Each encoder detects rotational position information of the corresponding motor and transmits it to the robot control device **9**. Based on the rotational position information of each motor received from each encoder, the robot control device **9** controls the energization conditions of each motor via a motor driver (not shown). As a result, the first arm **23**, the second arm **24**, and the spline shaft **253** are operated, and the posture of the robot arm **22** is changed over time according to a predetermined program, so that desired work can be performed.

[0032] Next, the structure of the second arm **24** will be described.

[0033] As shown in FIGS. **1** and **2**, the second arm **24** includes an arm base **3**, which forms the basic skeleton of the second arm **24**, and the cover **4**, which is responsible for most of the external

shape of the second arm **24**. The cover **4** functions as a casing for covering and protecting mounted object of the second arm **24**.

[0034] The arm base **3** is configured by a rigid body, and supports each part of the mounted objects mounted on the second arm **24**, such as the working head **25**, the second joint actuator **28**, the first drive mechanism **291**, and the second drive mechanism **292**. The arm base **3** is mainly configured by a plate-like member having a predetermined thickness in the Z-axis direction and having a linear shape, that is, a longitudinal shape extending in the X-axis direction. Further, a portion existing in a direction orthogonal to the longitudinal direction of the arm base **3**, that is, in the Y-axis direction is referred to as a side portion. The thickness of the arm base **3** is equal to the length of a side surface **30** of the arm base **3** in the Z-axis direction.

[0035] The arm base **3** may have a plate-like composition as a whole and include a hollow portion and ribs for partitioning the hollow portion into a plurality of spaces. Thus, the arm base **3** can achieve both securing of a sufficient strength and weight reduction. In the arm base **3** having such a configuration, the side surface **30** is formed on the outer surface of a side wall formed along the outer periphery of the arm base **3**.

[0036] Examples of the constituent material of the arm base **3** include various metal materials, various resin materials, in particular, a hard resin material, and various ceramics, and a composite material obtained by arbitrarily combining these materials may be used.

[0037] The arm base **3** includes an installation section **31** on which the second joint actuator **28** is installed and fixed, and an installation section **32** on which the working head **25** is installed and fixed.

[0038] The installation section **31** is provided at the base end portion of the arm base **3**, and is configured by a through hole penetrating in a thickness direction of the arm base **3**. An outer circumferential surface of an output shaft of the decelerator (not shown) of the second joint actuator **28** is fixed to an inner circumferential surface of the installation section **31**.

[0039] The installation section **32** is provided at the tip end portion of the arm base **3**, and is configured by a through hole penetrating in the thickness direction of the arm base **3**. An outer circumferential surface of a bearing (not shown) for supporting the spline shaft **253** of the working head **25** is fixed to an inner circumferential surface of the installation section **32**.

[0040] The cover **4** covers an upper portion of the arm base **3**, and has a function of protecting each section of the mounted object mounted inside the second arm **24**, such as the working head **25** and the second joint actuator **28**. The cover **4** is detachably and attachably mounted on the arm base **3**. In the present embodiment, when the cover **4** is mounted on the arm base **3**, the cover **4** is fixed via screws **5** as fixing members. This will be described in detail later.

[0041] The cover **4** includes a cover main body **41** and a covering section **42** for covering a part of the side surface **30** of the arm base **3**.

[0042] The cover main body **41** is configured by a housing and covers the upper portion of the arm base **3**. The cover main body **41** is configured by, for example, a plate material obtained by molding a resin material into a desired three dimensional shape. It is desirable that the cover main body has elasticity to such an extent that it is slightly deformed when a force is applied. Further, the cover main body **41** may be formed by molding a plate material made of a metal material such as stainless steel or aluminum into a desired three dimensional shape.

[0043] As shown in FIGS. **2**, **3**, **7**, and **8**, the cover main body **41** includes the top surface **41A** and a base end surface **41B**, and a slit **411** is formed in both the top surface **41A** and the base end surface **41B**. The slit **411** is formed along the X-axis direction in the top surface **41A**, and is formed from the vicinity of the center portion of the cover main body **41** in the X-axis direction to the base end side. The slit **411** is formed from the upper end to the lower end along the Z-axis direction in the base end surface **41B**. The slit **411** is formed from a portion where the pipe **10** penetrates the top surface **41A** to the vicinity of the center portion of the base end surface **41B** in the Z-axis direction, and includes a wide opening section **412** where the slit is wide. When the cover **4** is to be

detached from the arm base **3**, the pipe **10** moves along the longitudinal direction of the wide opening section **412**. Therefore, the width of the wide opening section **412**, that is, the length thereof in the Y-axis direction is substantially equal to the outer diameter of the pipe **10**.

[0044] The slit **411** and the wide opening section **412** are provided to apply an external force to the cover **4** to deform the cover **4** when the cover **4** is attached to or detached from the arm base **3**, thereby facilitating the attachment and the detachment operation. The shapes and arrangement patterns of the slit **411** and the wide opening section **412** are not limited to the configurations shown in the drawings.

[0045] The covering section **42** is provided to protrude downward from the cover main body **41**, and covers a part of the side surface **30** of the arm base **3** in a state in which the cover **4** is mounted on the arm base **3** (hereinafter, simply referred to as a “mounted state”). In the configuration shown in the drawings, it is provided approximately halfway around so as to cover a portion on the base end side among the side surface **30** of the arm base **3**. In other words, the covering section **42** is provided so as to cover a portion corresponding to the second joint actuator **28**, which is the joint section of the side surface **30** of the arm base **3**, that is, the outer circumferential portion of the second joint actuator **28**, and is not provided on the outer circumferential portion of the installation section **32** of the working head **25**. As a result, the rigidity of the portion of the cover **4** corresponding to the second joint actuator **28** can be further increased, and the effect of preventing the entry of dust, dirt, and the like into the vicinity of the second joint actuator **28** is also increased. Due to these synergistic effects, the second joint actuator **28** can be effectively protected. Further, it is possible to effectively prevent an operator from installing a peripheral device, wiring, or the like in the portion, and it is possible to prevent the operation of the robot arm **22** from being hindered by the installation.

[0046] A portion among the side surface **30** of the arm base **3** closer to the tip end side than the central portion is not covered with the covering section **42**, that is, is exposed from the covering section **42**. Hereinafter, the outer surface of the covering section **42** is referred to as a first surface **42A**, a portion among the side surface **30** exposed from the covering section **42** is referred to as a second surface **30B**, and a portion among the side surface **30** covered with the covering section **42** is referred to as a third surface **30C**.

[0047] The formation position of the covering section **42** is not limited to the above configuration, and for example, in contrast to the above, the covering section **42** may be provided so as to cover a portion on the tip end side among the side surface **30** of the arm base **3**, and a portion on the base end side among the side surface **30** may be exposed.

[0048] The lower end of the covering section **42** is positioned below the lower end in the thickness direction of the arm base **3**. That is, the covering section **42** covers the side surface **30** beyond the lower end of the arm base **3**. Accordingly, the effect of covering the side surface **30** is further improved, and the effect of preventing the entry of dust, dirt, and the like, that is, the effect of protecting the inside by the cover **4** is further enhanced. However, without being limited to this configuration, the lower end of the covering section **42** may be positioned at the same position as the lower end of the arm base **3** in the Z-axis direction, or may be positioned above the lower end of the arm base **3** and below the upper end of the arm base **3**. In a case where the lower end of the covering section **42** exceeds the lower end of the arm base **3** or is at the same position as the lower end of the arm base **3**, and a finger F is easily hooked on the lower end of the covering section **42** without the lower end of the arm base **3** being an obstacle, and thus the detachment operation is easily performed.

[0049] As shown in FIG. 2, the distance in the Z-axis direction between the top surface **41A** of the cover **4** and the lower end of the cover main body **41**, that is, the ratio $D2/D1$ between the length **D1** in the Z-axis direction of the cover main body **41** and the length **D2** in the Z-axis direction of the side surface **30** is not particularly limited, but is desirably 0.1 or more and 0.8 or less, and more desirably 0.3 or more and 0.6 or less. Thus, the arm base **3** which is heavier and stronger than the

cover main body **41** is shown, and the cover main body **41** which is lighter than the arm base **3** occupies a larger proportion of the external shape of the second arm **24**, so that a viewer can have a sense of stability and a quick impression, and the aesthetic property can be enhanced.

[0050] In the present embodiment, the top surface **41A** of the cover **4**, the lower end of the cover main body **41**, the lower end of the covering section **42**, and the lower end of the arm base **3** extends along a horizontal direction. However, without being limited to this configuration, they may be inclined with respect to the horizontal direction.

[0051] As shown in FIG. **4**, when the angle of the range in which the covering section **42** exists around the second rotation axis **J2** as viewed from the axial direction of the second rotation axis **J2** is defined as $\theta 1$, $\theta 1$ is not particularly limited, but is desirably 10° or more and 300° or less, more desirably 45° or more and 280° or less, and still more desirably 90° or more and 270° or less.

Accordingly, the covering section **42** is formed over a wide range, and it is possible to necessarily and sufficiently exhibit the effect obtained by installing the covering section **42**, that is, the improvement effect of the operability of the detachment operation and the protection effect of the mounted object of the arm base **3**, particularly, the second joint actuator **28**.

[0052] Note that, in the present embodiment, the covering section **42** is formed continuously in a circumferential direction within the range of the angle $\theta 1$, but it is not limited thereto, and a plurality of covering sections **42** may be formed intermittently or intermittently along the circumferential direction.

[0053] As shown in FIGS. **4** and **5**, the covering section **42** includes, on the base end portion side, a first portion **421**, which is not in contact with the side surface **30** of the arm base **3** and is spaced apart from the side surface **30** of the arm base **3** in the mounted state of the cover **4**. In the present embodiment, the first portion **421** has a stepped shape protruding toward the base end portion side of the second arm **24**, but may spread out downward, that is, may be inclined away from the side surface **30**. In this case, the first portion **421** may be referred to as a skirt portion. In addition, since the first portion **421** is particularly suitable for hooking a finger F when the cover **4** is gripped, it is also referred to as a finger hooking section.

[0054] By providing such a first portion **421**, when the cover **4** is gripped and detached from the arm base **3**, as shown in FIG. **5**, since the first portion **421** is separated from the side surface **30**, a finger F is unlikely to touch the arm base **3**, and a finger F easily hooks the first portion **421**. A finger F of the hand may be inserted between the first portion **421** and the side surface **30** and a finger F may hook the lower end of the first portion **421**. Therefore, it is easy to grip the cover **4**, it is easy to lift the cover **4**, and it is possible to easily and quickly perform the detachment operation of the cover **4** from the arm base **3** (hereinafter, simply referred to as the “detachment operation”). In the present embodiment, the base end portion side of the covering section **42** is the first portion **421**. By this, since the first portion **421** becomes closer to the slit **411** provided in the base end surface **41B**, the operator can easily open the slit **411** in the detachment operation (to be described later), and the detachment operation can be performed easily and quickly. Note that the entire covering section **42** may be the first portion **421**. That is, the first portion **421** may be formed over the entire circumference of the covering section **42**. In this case, the detachment operation can be easily performed by hooking a finger F at any portion of the covering section **42**.

[0055] The maximum separation distance D of the first portion **421** from the side surface **30** (third surface **30C**) is not particularly limited, but is desirably 10 mm or more and 100 mm or less, and more desirably 15 mm or more and 50 mm or less. Thus, by increasing the distance between the first portion **421** and the side surface **30**, a finger F is unlikely to contact the arm base **3**, and a finger F can be inserted into the gap without excess or deficiency, and the detachment operation can be easily performed. In the case where the first portion **421** has a shape inclined so as to spread downward, the lower end of the first portion **421** is at the maximum separation distance D from the side surface **30** (the third surface **30C**).

[0056] The first portion **421** may have a shape in which the separation distance with the side

surface **30** of the arm base **3** increases continuously or stepwise toward the lower end of the covering section **42**. That is, the first portion **421** may be provided to be inclined so as to spread downward. Thus, as shown in FIG. 5, when a finger F is inserted between the first portion **421** and the side surface **30** (the third surface **30C**), since the separation distance becomes narrower toward the upper side, the insertion depth of a finger F can be restricted, and the cover **4** can be more easily gripped.

[0057] When the separation distance between the first portion **421** and the side surface **30** (the third surface **30C**) of the arm base **3** continuously increases toward the lower end of the covering section **42**, an inclination angle $\theta 3$ of the first portion **421** with respect to the side surface **30** of the arm base **3** is not particularly limited, but is desirably 10° or more and 70° or less, and more desirably 15° or more and 45° or less. By this, it is possible to more reliably exhibit the above described effect, and it is possible to prevent the covering section **42** from becoming an obstacle when the cover **4** is attached to and detached from the arm base **3**.

[0058] The inclination angle $\theta 3$ is not limited to being constant from the upper end to the lower end of the first portion **421**, and may include a portion where the inclination angle $\theta 3$ changes at an intermediate location.

[0059] The present disclosure is not limited to the above described configuration, and the separation distance between the first portion **421** and the side surface **30** (the third surface **30C**) of the arm base **3** may increase stepwise toward the lower end of the covering section **42**.

[0060] Further, the covering section **42** may be provided in parallel with respect to the side surface **30** (the third surface **30C**) of the arm base **3** at a predetermined interval.

[0061] The separation distance between the first portion **421** and the third surface **30C**, in particular, the maximum separation distance D and the inclination angle $\theta 3$ may be constant along the circumferential direction of the covering section **42**, or may have partially different portions.

[0062] Further, the inner surface of the covering section **42** may be in surface contact with the side surface **30** (third surface **30C**) of the arm base **3**. That is, the first portion **421** may be omitted.

[0063] As shown in FIG. 4, when the angle of the range in which the first portion **421** exists around the second rotation axis J2 as viewed from the axial direction of the second rotation axis J2 is defined as $\theta 2$, $\theta 2$ is not particularly limited, but is desirably 5° or more and 300° or less, and more desirably 30° or more and 270° or less. By this, the first portion **421** is formed over a relatively wide range, and can be used for the detachment operation, so that the above described effects can be exhibited as necessary and as sufficient.

[0064] Note that, in the present embodiment, the first portion **421** is formed continuously in the circumferential direction within the range of the angle $\theta 2$, but it is not limited thereto, and a plurality of first portions **421** may be formed intermittently or intermittently along the circumferential direction.

[0065] When the relationship between $\theta 2$ and $\theta 1$ is considered, $\theta 2/\theta 1$ is desirably 0.1 or more and 1 or less, more desirably 0.2 or more and 0.8 or less, and still more desirably 0.3 or more and 0.6 or less. As a result, the range in which the first portion **421** exists can be secured in a necessary and sufficient manner, and the effect of preventing entry of dust, dirt, and the like, that is, the effect of protecting the inside by the cover **4**, and the effect of improving the operability of the detachment operation both can be achieved in a well-balanced manner. In the present embodiment, as shown in FIG. 4, $\theta 2/\theta 1=0.33$.

[0066] As shown in FIGS. 1 to 4, 6, 7, and 8, the cover **4** is fixed to the arm base **3** by the screws **5**, which are fixing members, in a state where the cover **4** is mounted on the arm base **3**. In particular, it is fixed to the side surface **30** of the arm base **3**. As a result, the cover **4** can be reliably and stably fixed with respect to the arm base **3**, and even when vibration, impact, other external force, or the like is applied during work, the cover **4** can be prevented from falling off. The screw **5** is inserted and screwed into a through hole (not shown) formed in the covering section **42** and a screw hole formed in the side surface **30** (third surface **30C**) of the arm base **3**, thereby fixing the covering

section **42** to the arm base **3**. Thus, the cover **4** can be fixed to the arm base **3** by a simple method of forming a screw hole in the side surface **30** (the third surface **30C**) of the arm base **3**.

[0067] In the shown configuration, as a configuration of screwing to the covering section **42** without the first portion **421**, the covering section **42** on the side portion of the arm base **3** is fastened by screw, but the covering section **42** with the first portion **421** may be fastened by screw. In the case where the first portion **421** is fastened by screw, fastening by screw can be easily and reliably performed as described above by using a relatively long screw **5** or by fastening the covering section **42** by screw at a position higher than the first portion **421**. In particular, it is desirable to perform fastening by screw as described above when the entire covering section **42** is the first portion **421**.

[0068] The portion to be fastened by the screw **5**, that is, the fixing portion, may be one or a plurality of portions, but a plurality of portions are desirable in order to make the fixing more reliable. In the present embodiment, the locations that are fastened by the screws **5** are a total of two places on the plus side in the Y-axis direction and the minus side in the Y-axis direction of the third surface **30C** among the side surface **30**, but the present disclosure is not limited thereto, and one or two or more places may be formed on the base end surface of the third surface **30C**, that is, the surface on the plus side in the X-axis direction.

[0069] Alternatively, only the covering section **42** may be fastened by screw to the arm base **3**, only the cover main body **41** may be fastened by screw to the arm base **3**, or both the cover main body **41** and the covering section **42** may be fastened by screw to the arm base **3**.

[0070] Although the case where the screw **5** is used as the fixing member has been described, the present disclosure is not limited thereto, and fixing members other than the screw **5**, for example, a bis, a bolt and a nut, a hook, a clip, a tightening band, a magnet, or the like may be used, or fixing may be performed by a method such as caulking or fitting.

[0071] Next, the procedure of the detachment operation for detaching the cover **4** from the arm base **3** will be described.

Procedure 1

[0072] In the mounted state of the cover **4**, first, all the screws **5** are removed using a screwdriver (not shown).

Procedure 2

[0073] Next, as shown in FIG. **6**, for example, the covering section **42** is gripped with the right hand HR, and the tip end portion of the cover main body **41** is gripped with the left hand HL. At this time, as shown in FIG. **5**, a finger F of the right hand HR is inserted into and hooks the inside of the first portion **421** of the covering section **42**.

Procedure 3

[0074] Next, the tip end portion of the cover main body **41** is lifted upward with the left hand HL, and the covering section **42** is lifted upward with the right hand HR while a finger F is hooked on the first portion **421**. When the cover **4** is lifted, the pipe **10** moves in the wide opening section **412** of the slit **411** in its longitudinal direction.

Procedure 4

[0075] Next, while maintaining the state in which the cover **4** is lifted in the procedure 3, as shown in FIGS. **7** and **8**, the left hand HL is changed to grip the portion of the covering section **42** on the Y-axis minus side with respect to the slit **411**, and the right hand HR is changed to grip the portion of the covering section **42** on the Y-axis plus side with respect to the slit **411**. At this time, as shown in FIG. **7**, the left hand HL may grip the side portion on the Y-axis minus side of the covering section **42**, and the right hand HR may grip the side portion on the Y-axis plus side of the covering section **42**, or as shown in FIG. **8**, the left hand HL may grip the first portion **421** on the Y-axis minus side of the slit **411**, which is the base end portion side of the covering section **42**, and the right hand HR may grip the first portion **421** on the Y-axis plus side. Then, a force is applied to the cover main body **41** so that the portion of the cover main body **41** on the minus side of the Y axis

and the portion on the plus side of the Y axis separate from each other, that is, the slit **411** is opened, and in the state where the slit **411** is open, the cover **4** is slightly lifted so that the pipe **10** is detached from the slit **411**.

Procedure 5

[0076] Next, the cover **4** is further lifted upward so as to remove the spline shaft **253** from the cover **4**.

[0077] The cover **4** can be detached from the arm base **3** through such procedures 1 to 5. In the procedures 1 to 5, the use of left and right hands may be reversed.

[0078] Here, when the covering section **42** is omitted, it is difficult to hold the lower end of the cover **4**, or it is difficult to know where to hold. Since the covering section **42** is provided in the robot **2**, the operator can easily recognize the covering section **42**, particularly the first portion **421**, which is a portion convenient for hooking a finger F, and can easily hook a finger F on the first portion **421**. As a result, the operator can stably grip the cover **4**. Therefore, the cover **4** can be easily and quickly detachment operation. After the cover **4** is detached, maintenance such as inspection, repair, and replacement of parts can be performed on the mounted object on the arm base **3** and the inside of the cover **4**.

[0079] As described above, the robot **2** includes the base **21**, the first arm **23** rotatably connected to the base **21** around the first rotation axis **J1**; and the second arm **24** rotatably connected to the first arm **23** around the second rotation axis **J2** parallel to the first rotation axis **J1**, wherein the second arm **24** includes the arm base **3** and the cover **4** configured to attach to and detach from the arm base **3** and the cover **4** includes the cover main body **41** that covers the upper portion of the arm base **3** and the covering section **42** that protrudes downward from the cover main body **41** and that covers a portion of the side surface **30** of the arm base **3**.

[0080] Accordingly, when the cover **4** is detached from the arm base **3**, the cover **4** can be easily gripped, and thus a detachment operation of the cover **4** can be easily and quickly performed.

[0081] Also, the robot **2**, further includes the screw **5** as a fixing member for fixing the covering section **42** to the side surface **30** of the arm base **3**.

[0082] Thus, the cover **4** can be reliably and stably fixed with respect to the arm base **3**.

[0083] Also, the robot **2**, further includes the second joint actuator **28** that is the joint section that rotatably connects the second arm **24** to the first arm **23** around the second rotation axis **J2**, wherein the covering section **42** covers the portion of the side surface **30** of the arm base **3** corresponding to the second joint actuator **28**.

[0084] As a result, the rigidity of the portion of the cover **4** corresponding to the second joint actuator **28** can be increased, and the effect of preventing the entry of dust, dirt, and the like into the vicinity of the second joint actuator **28** is also increased. Due to these synergistic effects, the second joint actuator **28** can be effectively protected.

[0085] In addition, the covering section **42** includes the first portion **421** separated from the side surface **30** of the arm base **3**.

[0086] Thus, when the cover **4** is detached from the arm base **3**, as shown in FIG. 5, a finger F can be inserted between the first portion **421** and the side surface **30**, and a finger F can be hooked on the lower end of the first portion **421**. Therefore, the cover **4** be easily gripped, and can be easily and quickly detachment operation.

[0087] In addition, the first portion **421** has a shape in which the separation distance from the side surface **30** of the arm base **3** increases toward the lower end of the covering section **42**.

[0088] Thus, as shown in FIG. 5, when a finger F is inserted between the first portion **421** and the side surface **30** (the third surface **30C**), since the clearance on the upper side is narrowed, the insertion depth of a finger F can be restricted, and the cover **4** can be easily gripped. Therefore, the cover **4** can be detached more easily and quickly.

[0089] As shown in FIG. 4, $\theta 2/\theta 1$ is equal to or greater than 0.1 and equal to or less than 1, when an angle of the range in which the covering section **42** exists around the second rotation axis **J2** is

$\theta 1$, and an angle of the range in which the first portion **421** exists around the second rotation axis **J2** is $\theta 2$, as viewed from the axial direction of the second rotation axis **J2**.

[0090] Accordingly, it is possible to secure a necessary and sufficient range in which the first portion **421** is present, and it is possible to more reliably exhibit the above described effect. In particular, both the internal protection effect by the cover **4** and the operability improvement effect of the detachment operation can be achieved in a well-balanced manner.

[0091] In such a robot **2**, the first surface **42A**, which is the outer surface of the covering section **42**, and the second surface **30B**, which, among the side surface **30** of the arm base **3**, is not covered with the covering section **42**, are different from each other in at least one of the wavelength and intensity of reflection light obtained by reflecting the visible lights incident thereon. Accordingly, the distinguishability of the boundary between the first surface **42A** and the second surface **30B** is secured, and the arm base **3** and the covering section **42** can be easily visually distinguished from each other. Therefore, when the cover **4** to be detached from the arm base **3**, the covering section **42** is easily recognized. As a result, the cover **4** can be easily and quickly detached by immediately holding the covering section **42** and performing the above described operation. This will be described in detail below.

[0092] Among the wavelength and the intensity of the reflection light on the first surface **42A** and the second surface **30B** will be described, a case where the wavelength are mainly different will be described as a case 1, a case where the intensity are mainly different will be described as a case 2, and a case where both the wavelength and the intensity are different as a case 3.

Case 1

[0093] For the case where the wavelengths of reflection light mainly differ between the first surface **42A** and the second surface **30B**, an example of at least one of hue, saturation, and lightness differing between the first surface **42A** and the second surface **30B** can be raised.

[0094] In any of the case where hues are different, the case where saturations are different, and the case where lightness are different, the above described effect can be obtained as long as a color difference ΔE is different between the first surface **42A** and the second surface **30B** when the reflection light is measured using a color difference meter. The color difference ΔE is desirably 5.0 or more, and more desirably 12.0 or more. As a result, the above mentioned effect can be exhibited more reliably.

[0095] Both of the first surface **42A** and the second surface **30B** may be chromatic color, the first surface **42A** may be chromatic color and the second surface **30B** may be achromatic color, the first surface **42A** may be achromatic color and the second surface **30B** may be chromatic color, or both of the first surface **42A** and the second surface **30B** may be achromatic color.

[0096] The case where both the first surface **42A** and the second surface **30B** are chromatic color, the case where the first surface **42A** is a chromatic color and the second surface **30B** is an achromatic color, and the case where the first surface **42A** is an achromatic color and the second surface **30B** is a chromatic color, the design of the entire robot **2** can be made gorgeous, and can be made relatively conspicuous to give an impression of presence.

[0097] In a case where both the first surface **42A** and the second surface **30B** are achromatic color, the design of the entire robot **2** is monotone and can be made chic, and can be made to have a quiet color tone that is not too conspicuous.

[0098] When both the first surface **42A** and the second surface **30B** are achromatic color, only the lightness is different. In this case, the difference in Munsell lightness value measured with a lightness meter is desirably 2 or more, and more desirably 4 or more. As a result, the above mentioned effect can be exhibited more remarkably.

[0099] Such the case 1 is produced by a method in which constituent materials having different colors are used for the arm base **3** and the covering section **42**, a method in which a dye or a pigment is blended into at least one of the constituent material itself of the arm base **3** or the constituent material itself of the covering section **42**, and coloring is performed, a method in which

a coating material such as a coloring coating material is applied to at least one of the first surface **42A** and the second surface **30B**, a method in which a film such as a coloring film or a colored transparent film is adhered to at least one of the first surface **42A** and the second surface **30B**, or a method in which these methods are arbitrarily combined.

[0100] The constituent material of the arm base **3**, the constituent material of the covering section **42**, the paint, and the film may contain a fluorescent substance, a luminescent substance, metal powder, or the like.

Case 2

[0101] For the case where the intensities of reflection light mainly differ between the first surface **42A** and the second surface **30B**, an example of hue, saturation, and lightness being substantially the same and glossiness of the first surface **42A** and the second surface **30B** differing can be raised.

[0102] The first surface **42A** may have a higher glossiness than the second surface **30B**, or the second surface **30B** may have a higher glossiness than the first surface **42A**.

[0103] The glossiness can be measured with a glossmeter.

[0104] As a method of making the glossiness different between the first surface **42A** and the second surface **30B**, a method of making surface roughness different, a method of making visible light absorptivity of the constituent material different, and the like can be raised as examples.

[0105] Such a case 2 can be realized by a method in which the surface of either one of the first surface **42A** and the second surface **30B** is roughened or embossed by sand blast, shot blast, satin finish, hairline finish, chemical treatment, or the like, a method in which the surface of either one of the first surface **42A** and the second surface **30B** is polished, glossed, or mirror finish, a method in which the surface of either one of the first surface **42A** and the second surface **30B** is clear-coated or matte-coated, a method in which the surface of the first surface **42A** and the second surface **30B** is light-transmitting or light-non-transmitting films having different degrees of glossiness are attached, a method in which these methods are arbitrarily combined, or the like.

Case 3

[0106] For the case where the both the wavelength and the intensity of the reflection light are different between the first surface **42A** and the second surface **30B**, an example of at least one of hue, saturation, and lightness differ and the glossiness differ between the first surface **42A** and the second surface **30B** can be raised. That is, case 3 is a combination of case 1 and case 2.

[0107] Case 3 includes a case where the first surface **42A** is metallic color and the second surface **30B** is chromatic color, a case where the first surface **42A** is metallic color and the second surface **30B** is achromatic color, a case where the first surface **42A** is chromatic color and the second surface **30B** is metallic color, and a case where the first surface **42A** is achromatic color and the second surface **30B** is metallic color. Further, a case where the first surface **42A** and the second surface **30B** have different metallic colors is also included.

[0108] Examples of the metallic color include gold, silver, copper, and pink gold. In the case of the metallic color, there is an advantage that it is easy to vary the intensity of the reflection light, that is, the glossiness by setting the property of the front face such as the surface roughness.

[0109] As described above, since the first surface **42A** and the second surface **30B** are different from each other in at least one of the wavelength and the intensity of the reflection light, the arm base **3** and the covering section **42** can be visually easily distinguished from each other. Therefore, at the time of the detachment operation of the cover **4**, the covering section **42** (the first surface **42A**) can be distinguished and easily visually recognized separately from the side surface **30** (the second surface **30B**) of the arm base **3**. That is, first, when the fixation of the cover **4** is released, it is possible to easily visually recognize the screwed portion by the screw **5** formed in the covering section **42**, and then, when the cover **4** is gripped, it is possible to easily visually recognize the covering section **42** which is a portion where a finger **F** is hooked. In addition, the cover **4** can be detached more easily and quickly by a synergistic effect with the effect of providing the covering section **42** as described above.

[0110] Further, among the side surface **30** of the arm base **3**, the second surface **30B** which is not covered with the covering section **42** and the third surface **30C** which is covered with the covering section **42** desirably have the same wavelength and intensity of reflection light. Since the third surface **30C** is covered with the covering section **42** and cannot be seen, there is no difference in the distinguishability of the boundary between the first surface **42A** and the second surface **30B** even if the second surface **30B** and the third surface **30C** have the same color or different colors. Further, by making the second surface **30B** and the third surface **30C** have the same color, that is, the same wavelength and intensity of the reflection light, the entire side surface **30** of the arm base **3** can be integrally configured in the same color, the configuration of the arm base **3** can be simplified, and the arm base **3** can be easily manufactured.

[0111] Note that the first surface **42A** and the second surface **30B** are not limited to the case of a single color, and may be provided with a marker or a pattern indicating the boundary or the portion to be gripped. In addition, the first surface **42A** and the second surface **30B** may have different tactile sensation and thermal conductivity. In this case, for example, even in a relatively dark room, it is possible to easily, quickly, and reliably grasp the portion of the cover **4** to be gripped.

[0112] Hereinabove, the robot according to the disclosure has been described based on the embodiments shown in the drawings, but the disclosure is not limited thereto, and the configuration of each section can be replaced with an arbitrary configuration having the same function. Further, the robot according to the present disclosure may be provided with other arbitrary components.

Claims

1. A robot comprising: a base; a first arm rotatably connected to the base and rotatable around a first rotation axis; and a second arm rotatably connected to the first arm and rotatable around a second rotation axis parallel to the first rotation axis, the second arm extending along a first direction, the second arm having first and second ends opposite to each other along the first direction, wherein the second arm includes a working head, an arm base, and a cover configured to attach to and detach from the arm base, the working head is disposed at a first position closer to the first end than the second end of the second arm, the second rotation axis is located at a second position closer to the second end than the first end of the second arm, the cover includes: a cover main body that covers an upper portion of the second arm; and a covering section that downwardly continuously extends from the cover main body and that covers a side surface of the arm base at the second end of the second arm, a bottom of the covering section downwardly extends further than a bottom of the cover main body at the first end of the second arm, and the side surface of the arm base at the second end of the second arm is spaced apart from the covering section along the first direction without having another structure therebetween.
2. The robot according to claim 1, further comprising: a fixing member for fixing the covering section to the side surface of the arm base.
3. The robot according to claim 1, further comprising: a joint section that rotatably connects the second arm and the first arm to rotate the second arm around the second rotation axis, wherein the covering section covers a portion of the joint section.
4. The robot according to claim 1, wherein the covering section includes a first portion separated from the side surface of the arm base at the second end of the second arm.
5. The robot according to claim 4, wherein a top part of the first portion is spaced apart from the side surface of the arm base by a first distance, and a bottom part of the first portion is spaced apart from the side surface of the arm base by a second distance, and the second distance is larger than the first distance.
6. The robot according to claim 4, wherein θ_2/θ_1 is in a range of 0.1 to 1, where, as viewed from an axial direction of the second rotation axis, θ_1 is an angle range in which the covering section exists

around the second rotation axis and θ_2 is an angle range in which the first portion exists around the second rotation axis.
